RESEARCH SHARING

PO: Value-Model-Based RL for Advanced Reason

Achieves 60.4 on AIME 2024 — 10+ Points Above DAPO

"Overcoming Core Challenges in Long Chain-of-Thought"

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Value-Free (e.g., GRPO, DAPO)

- Computes advantage based on final trajectory reward
- Assigns reward uniformly across all tokens
- High variance in credit assignment
- Suboptimal for long Chain-of-Thought where subtle errors cause catastrophic failures

Value-Based (VAPO)

- Precise token-level credit assignment
- Traces each action's specific impact on returns
- Lower-variance estimates per token
- Better generalization across samples
- Superior sample efficiency

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1. Value Model Bias

Accumulates over long trajectories.

Bootstrapped learning exacerbates initial errors, causing RL instability.

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2. Variable Sequence Lengths

Heterogeneous lengths require different

bias-variance tradeoffs. A fixed GAE λ parameter fails to adapt effectively.

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3. Reward Signal Sparsity

Only the final <eos> token receives an explicit reward.

Makes learning correct intermediate steps extremely difficult in long-CoT.

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Value-Pretraining → Solves: Reduces initial value model bias before RL

Length-Adaptive GAE → Solves: Dynamic λ adjusts bias-variance for lengths

Decoupled-GAE → Solves: Separates computation for robust value estimation

Additional Enhancements: Clip-Higher | Token-level Loss | Positive Example LM Loss | Group-Sampling

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A novel technique optimizing the bias-variance tradeoff across heterogeneous sequence lengths.

Short Responses

Action: Uses **Lower λ**

Effect:

Why:

as much from accumulated bias.

Long Responses

Action: Uses **Higher λ**

Effect:

Why:

accumulation in long reasoning.

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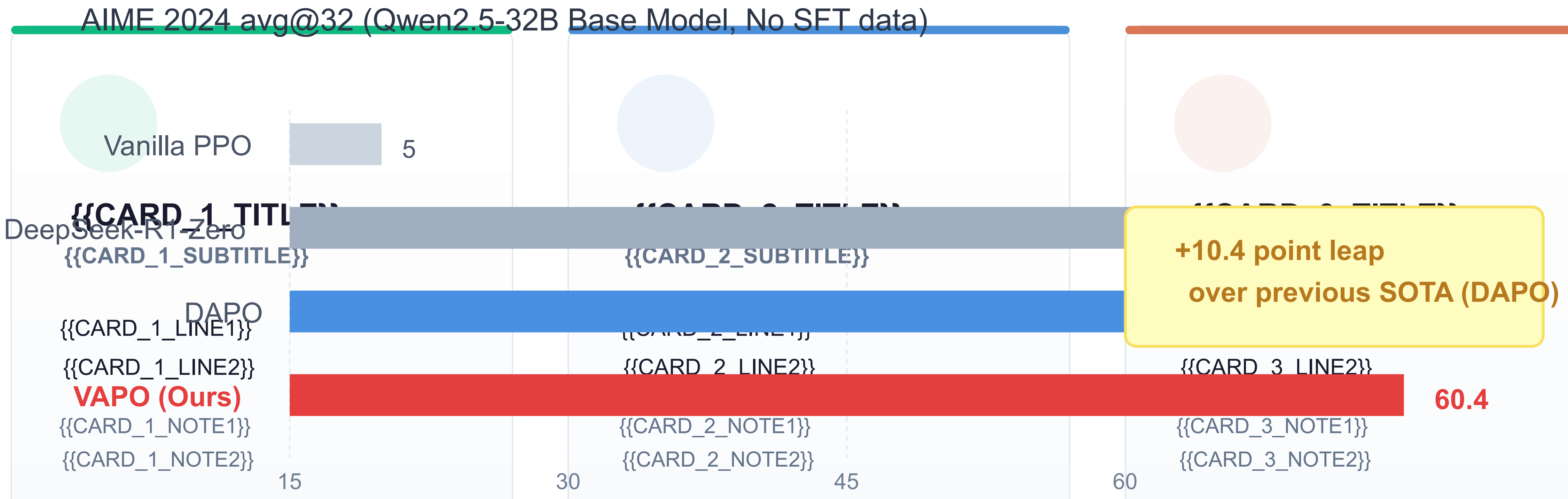

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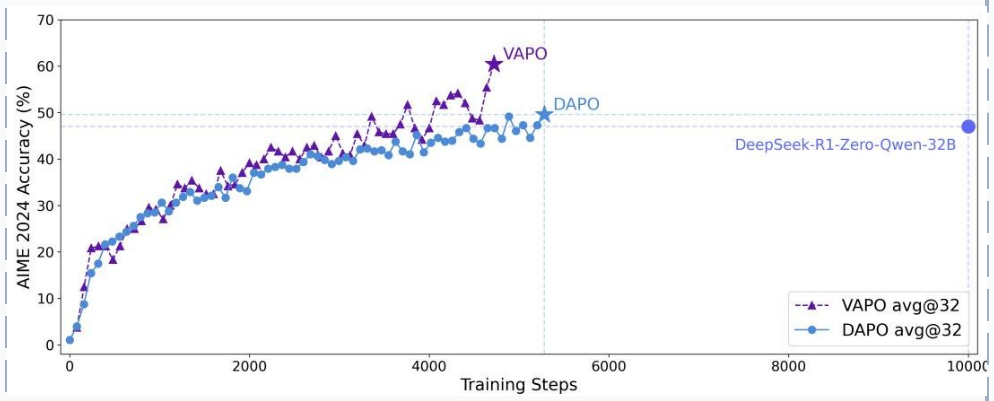
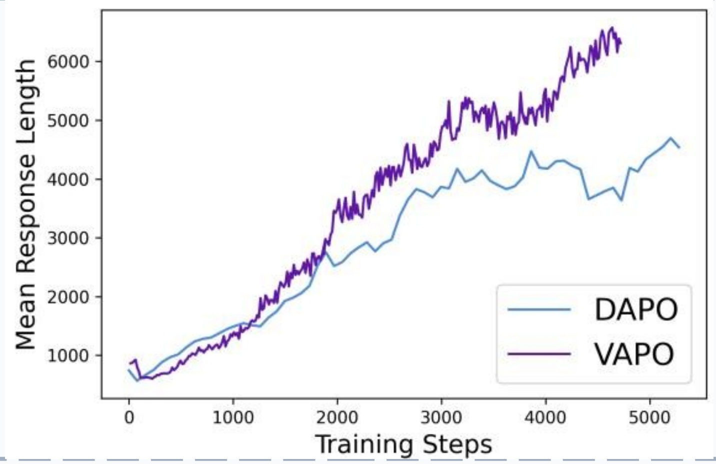
Highly efficient and stable: Zero training crashes across runs

AIME Accuracy over Training Steps



Response Length over Training Steps



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• VAPO hits 60% accuracy around 4,500 steps, while DAPO plateaus near 50% at 5,500 steps.

• Generates longer, more elaborate reasoning chains (~6,000 tokens vs DAPO's ~4,500).

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Model Configuration	AIME24 Score	Impact / Drop
Full VAPO Framework	60	-
w/o Value-Pretraining	**11**	**↓ 49 pts (Critical)**
w/o Decoupled-GAE	33	↓ 27 pts
w/o Length-adaptive GAE	45	↓ 15 pts
w/o Clip-Higher	46	↓ 14 pts

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Insight: Without Value-Pretraining, value-model-based RL barely functions (drops from 60 to 11).

THANK YOU

Thank You

1. Value-based methods can surpass value-free when structural challenges are systematically addressed.

2. Length-adaptive GAE is critical for variable reasoning lengths.

3. Value-pretraining is the single most important component.

4. VAPO is highly stable, efficient, and crash-free.

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